



An Hung Nguyen

COMPUTER SCIENTIST / AI RESEARCHER

Danang, Vietnam

+84354430933 | nguyenanhung1903@gmail.com | 📅 March 19th, 2003 | 🏠 nguyenhung.me | github.com/nguyenhung1903

🌐 linkedin.com/in/nguyenhung1903 | k.nguyenhung1903 | 🎓 An Hung Nguyen

Personal Profile

I am a researcher with a strong passion for Artificial Intelligence and Data Science. I have applied AI to real-world problems in medicine and agriculture through collaborations with scientific labs in Vietnam and internationally.

My research interests include Computational Biology, Bioinformatics, Mathematics, Hyper-complex and Quaternion numbers, View Synthesis, Neural Rendering, and Real-Time Graphics.

Education

University of Science and Technology - The University of Danang

Danang, VN

B.S., Information Technology

2021 - 2025

- **Grade:** 3.65/4.0
- Excellent Bachelor's Degree
- Working as an AI Researcher with Minh Tuan Pham PhD.
- **Capstone Project:** A Deep Learning-Based Approach for Multi-Weather Image Restoration and Enhancement. (<http://thuvienso.dut.udn.vn/handle/DUT/16775>)
- **Courses:** Data Science, Programming for Data Science, Knowledge Representation and Reasoning, Statistical Theory and Methods, Learning Skills through Case Studies, Artificial Intelligence, Machine Learning, Statistical Learning

Le Quy Don upper-secondary school for the gifted

Quang Tri, VN

Informatics class (IT 18-21)

2018 - 2021

- Specialised in Biology, Informatics, and Mathematics with Computer Science
- Selected as a member of the national-level scientific research competition team and the Informatics gifted student team.

Work Experience

SLA Software Logistik Artland GmbH

Germany (Remote)

Computer Scientist & AI Engineer

Aug 2024 - Present

- Support of integration of AI models into existing customer infrastructure.
- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.
- **Technical Skills:** Python with PyTorch, NumPy, Matplotlib, Pandas, Scikit-learn, C++, Ubuntu Linux, Linux tools, Apt, Scripting, Git.
- **Soft Skills:** Teamwork, Time Management, Communication, Presentation skills.
- **Research Skills:** Medical Research, Data Analysis, Data Engineering.

University of Regina

Canada

Collaboration Researcher

Aug 2024 - Sept 2025

- Researched solutions for simulating chemical processes.
- Implemented simulation code using Matlab and Python.

Phyigital Labs

Vietnam

AI Intern

May 2024 - Jul 2024

- Worked as an AI intern on deep learning projects.
- Worked with LLM models to propose solutions for querying large documents.
- Collaborated with a former Senior Google engineer to deploy systems.

FX Data

Germany (Remote)

Computer Scientist & AI Engineer

Jan 2022 - Jul 2024

- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.
- **Technical Skills:** Python with PyTorch, NumPy, Matplotlib, Pandas, Scikit-learn, C++, Ubuntu Linux, Linux tools, Apt, Scripting, Git.
- **Soft Skills:** Teamwork, Time Management, Communication, Presentation skills.
- **Research Skills:** Medical Research, Data Analysis, Data Engineering.

Rhen Consulting

Germany (Remote)

AI Engineer

Oct 2021 - Dec 2021

- Collaborated with an AI team to develop neural network architectures to detect and segment animal pathogens.
- Developed scripts and applications to automate and handle data.
- Automated and optimized the data handling process for signal processing, worked with Linux distributions, performed shell scripting, and utilized various Linux tools.

Highlight Awards

Fourth Place Award in the Field of Science, Engineering, and Technology

Vietnam

Vietnam Student Science Research and Innovation 2022 - 2023

Dec 2023

Project title: Quaternion Convolutional Neural Networks for Image Depth Estimation

Fourth Place Award in the in Computational Biology and Bioinformatics

Vietnam

Vietnam National Secondary Student Science and Engineering Fair

March 2021

Project title: Applying Deep Learning to the process of diagnosing, analyzing and detecting damage to the retina due to diabetes

Third Prize in the Provincial Contest for Excellent Students in Informatics

Vietnam

Quang Tri Department of Education and Training

Jan 2020

Publications

All-in-One Image Restoration with Quaternion Cross-Attention and Haar Frequency Decomposition

Submitted to Signal, Image and Video Processing (Journal Q2)

2025

- Status: Under review
- **An Hung Nguyen**, Minh Tuan Pham
- [dx.doi.org/10.21203/rs.3.rs-7260641/v1](https://doi.org/10.21203/rs.3.rs-7260641/v1)
- This work is part of my undergraduate research project and capstone thesis.

Optimizing Retrieval in Vietnamese Literary QA Using Named-Entity Relationship Graphs

Submitted to ACM Transactions on Asian and Low-Resource Language Information Processing (Q2)

2025

- Status: Under review
- Minh Tuan Pham, **An Hung Nguyen**, Phuc Hao Do, Thanh Huy Nguyen, and Nam Do

Real-Time 3D Reconstruction through Image Synthesis and IoT Integration

Manuscript under review

2025

- Cong Thang PHAM, **An Hung Nguyen**, Tan Sang Truong, Dai Quy Tran, Si Tien Tran
- Submitted to Vestnik St. Petersburg University: Mathematics

Adaptive Hybrid Transformer Network with Frequency Attention for All-in-One Image Restoration

Moscow, RUSSIA

PSBB25

June 09 - 11, 2025

- Cong Thang Pham, **An Hung Nguyen**, Quoc Cuong Nguyen, Minh Nhat Phan, Thanh Than Nguyen
- [10.5194/isprs-archives-XLVIII-2-W9-2025-213-2025](https://arxiv.org/abs/2506.05194)

DehazeCLNet: A Contrastive Learning Framework with Advanced Feature Extraction for Image Dehazing

Danang, Vietnam

SOICT 2024 (Springer CCIS)

December 13-15, 2024

- Cong Thang Pham, **An Hung Nguyen**, Quoc Cuong Nguyen, Minh Nhat Phan
- [10.1007/978-981-96-4282-3_23](https://doi.org/10.1007/978-981-96-4282-3_23)

Enhancing Image Classification Capabilities In The Vision Transformer Network Model With Quaternion Algebra

Danang, Vietnam

CITA 2024

July, 2024

- Minh Tuan Pham, **An Hung Nguyen**
- <https://elib.vku.udn.vn/handle/123456789/4031>
- <https://nguyenhung.me/paper/QViT.pdf>

Enhancing Convolutional Neural Network Architectures with Long Short-Term Memory for Improved Image Classification

Danang, Vietnam

IEEE: Institute of Electrical and Electronics Engineers

November 10-11, 2023

- **An Hung Nguyen**, Minh Tuan Pham
- 10.1109/ATIGB59969.2023.10364368

Quaternion Convolutional Neural Networks for Depth Estimation

IEEE: Institute of Electrical and Electronics Engineers

Mie, Japan

January 4, 2023

- **An Hung Nguyen**, Cao Duy Hoang, Dang Hoang Phu Phan, Minh Tuan Pham
- <https://doi.org/10.1109/SCISIS55246.2022.10002074>

Weight Initialization for Complex Number Convolutional Neural Networks Using Genetic Algorithms

Vietnam

Journal on Information Technologies and Communications

June 30, 2022

- Minh Tuan Pham, **An Hung Nguyen**
- <https://doi.org/10.32913/mic-ict-research-vn.v2022.n1.1054>

Efficient Complex Valued Neural Network with Fourier Transform on Image Denoising

Dubai, UAE

ACM: Association for Computing Machinery

December 16, 2021

- Minh Tuan Pham, Van Quang Nguyen, Cao Duy Hoang, **An Hung Nguyen**
- <https://doi.org/10.1145/3508072.3508081>

Skills

Programming	C/C++, Python (Pandas, PyTorch, NumPy, Scikit-learn. etc), Java, dot NET, React Native, SQL.
Miscellaneous	Linux, Shell (Bash/Zsh), $\text{\LaTeX}$ (Overleaf/R Markdown), Microsoft Office, Git, Arduino, MCU microcontroller.
Soft Skills	Time Management, Teamwork, Leadership, Problem-solving, Documentation, Engaging Presentation.

Top honors and awards

Mar 2025	2nd prizes , Faculty level Student Scientific Research and Innovation	Vietnam
Dec 2023	Third prize , University-level Student Scientific Research and Innovation	Vietnam
Dec 2023	First prize , Faculty level Student Scientific Research and Innovation	Vietnam
Dec 2023	4th prizes , Vietnam Student Science Research and Innovation	Vietnam
Jun 2023	3rd prizes , Faculty level Student Scientific Research and Innovation	Vietnam
Mar 2022	2nd prizes , Faculty level Student Science Research and Innovation	Vietnam
Aug 2021	Second prize (ranked 1st) , Innovative Generation 2021	Vietnam
Mar 2021	4th prizes , Vietnam National Secondary Student Science and Engineering Fair	Vietnam
Jan 2021	4th prizes , Provincial Contest for Excellent Students in Informatics	Vietnam
Jan 2020	3rd prizes , Provincial Contest for Excellent Students in Informatics	Vietnam

Languages

Chinese	Minimum Professional Proficiency
English	Minimum Professional Proficiency
Vietnamese	Native proficiency